

Empowering Non-Expert Users in Fashion Remanufacturing: Enhancing Human-Multi-Robot Interaction through Real-Time Visualization

Emanuel Gollob*
Creative Robotics
University of Arts Linz
Linz, Austria
emanuel@emanuelgollob.com

Johannes Braumann
Creative Robotics
University of Arts Linz
Linz, Austria
johannes@robotsinarchitecture.org

Amir Bastan*
Creative Robotics
University of Arts Linz
Linz, Austria
ping@amirbastan.com

Christiane Luible-Baer
Fashion & Technology
University of Arts Linz
Linz, Austria
christiane.luible-baer@kunstuni-linz.at

Abstract

Automating textile repair and remanufacturing can significantly improve the ecological and societal impact of textiles. However, the complex behavior of textiles poses challenges for integrating robotics into fashion. This research addresses these challenges by developing a platform that empowers non-experts to define complex multi-robot tasks through visualization-enhanced hand guiding. By leveraging collaborative automation, our approach facilitates intuitive human-multi-robot interaction without extensive technical knowledge. Our approach employs flexible, parametric systems and a dual-robot setup to achieve high precision and operational flexibility in remanufacturing processes. Utilizing Grasshopper for path planning and VVVV for real-time interaction, we create a user-friendly interface allowing non-experts to interact intuitively with robots. This work demonstrates how real-time visualization can make advanced robotic capabilities accessible to non-expert users in the fashion industry. By enabling non-experts to leverage these technologies, we aim to transform remanufacturing processes and foster innovation in human-robot collaboration in the fashion and textiles sector.

CCS Concepts

• **Computer systems organization** → **External interfaces for robotics**; • **Human-centered computing** → **Collaborative interaction**; • **Social and professional topics** → **Sustainability**.

Keywords

Collaborative Remanufacturing, Sustainable Fashion Industry, Human-Multi-Robot Interaction, Real-time Visualization

*Both authors contributed equally to this research.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '25, Yokohama, Japan

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3716232>

ACM Reference Format:

Emanuel Gollob, Amir Bastan, Johannes Braumann, and Christiane Luible-Baer. 2025. Empowering Non-Expert Users in Fashion Remanufacturing: Enhancing Human-Multi-Robot Interaction through Real-Time Visualization. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 9 pages. <https://doi.org/10.1145/3706599.3716232>

1 Introduction

From an EU consumption perspective, the supply chains of clothing, footwear, and household textiles are the 4th highest pressure category regarding the use of primary raw materials and water, 2nd highest for land use and 5th highest in terms of greenhouse gas emissions [1, 18]. As the production stages have the highest environmental impact, besides transitioning to circularity, the focus should also be on keeping fibers, textiles, and garments at their highest value during use and re-entering phases [22, 32]. Remanufacturing is a circular strategy that involves the process of recovering both raw materials and value from end-of-life products for the production of new items [6]. Compared to other industries where remanufacturing is widely adapted (e.g. electronics), it is underexplored in the textile and apparel industry [16]. Remanufacturing and repair of garments is currently done by hand by highly skilled craft persons. Calculating the amount of repair and remanufacturing necessary to meet the EU sustainability goals, further automation-supported methods have to be researched [2, 14, 28]. Therefore, our research investigates ways to keep apparel in the cycle longer by technologically empowering non-experts in fashion remanufacturing. To do this, this research implements a non-expert-centric approach that combines collaborative robotics, parametric robot control, and innovative apparel remanufacturing technologies.

The work presented here uses real-time animation and visualization to assist non-experts in defining complex robotic tasks in the field of fashion and textiles. The integration of robotics into the fashion industry is challenging due to the complex behavior of textiles, which makes it difficult to automate existing manual processes. Textiles are flexible, deformable, and vary in texture and thickness, which complicates robotic handling [19, 38]. Our approach is to develop a demonstrator that rethinks these processes

using collaborative automation and innovative technologies. This proposal emphasizes human-multi-robot interaction facilitated by real-time visualization.

A key challenge in a non-expert multi-robotic setup is to ensure that users, who may not have robotics expertise, can effectively communicate and operate the robotic system while interacting with two collaborative robots. The advantage of collaborative robots hereby is that they can be moved and taught by hand guiding, eliminating the need for programming. Robots that are teachable by hand guiding serve as both interfaces and production tools. To keep users informed of the process state, we use a visual workspace that displays a digital twin of the robot workspace, user input data, and planned toolpaths for the remanufacturing process. Real-time visualization is critical because it allows users to immediately see the impact of their actions on the overall process, ensuring seamless interaction and effective operation [8]. This user-friendly interface is designed to enable non-experts to use advanced robotic capabilities without requiring extensive technical knowledge (Figure 1).



Figure 1: Human-multi-robot setup for individualized apparel (re)manufacturing.

This paper presents the interaction design and technological aspects of developing such a collaborative multi-robot garment remanufacturing workflow and demonstrating it in an exhibition context.

2 Theoretical Background

The integration of advanced technologies in the fashion industry has opened new avenues for improving productivity, creativity, and sustainability. One such area is fashion remanufacturing, where the use of robotics and real-time visualization can significantly empower non-expert users and reduce textile waste. This chapter explores the theoretical underpinnings of empowering non-expert users in fashion remanufacturing, focusing on the role of human-multi-robot interaction and real-time visualization.

In the sustainable fashion discourse, repair is understood as the process of restoring the functionality and appearance of a garment through conventional techniques - such as mending or patching with similar fabric - that aim to preserve the original design. In contrast, remanufacturing extends beyond functional repair by

reclaiming both the raw materials and the intrinsic value embedded in end-of-life items and transforming them into new ones [6, 20]. This distinction is crucial: while repair seeks to return a garment to its previous state, remanufacturing embraces new garment designs based on end-of-life items. For example, a customized application of a new layer of fibers can not only prolong the lifecycle of a garment but also open avenues for reconfiguring the aesthetic, material and functional identity of the apparel. To effectively realize such remanufacturing with collaborative automation, it is vital to cultivate a fluid interaction between humans and robots.

Human-multi-robot interaction (HMRI) refers to the dynamic interplay between humans and multiple robots working collaboratively to achieve a common goal. This interaction is critical in scenarios where complex tasks require the coordination of multiple robotic agents, such as the remanufacturing of entire garments. The primary goal of HMRI is to increase the efficiency and effectiveness of human-robot teams by leveraging the strengths of both humans and robots. Therefore, effective task allocation is essential in HMRI to ensure that tasks are optimally distributed between robots and human collaborators. This involves taking into account the capabilities of each robot and the expertise of the human users to maximize overall performance [33, 40]. Equally critical to successful HMRI is seamless communication and coordination. This includes both verbal and non-verbal communication methods, such as gestures and eye gaze, to facilitate intuitive human-robot interactions [26, 60]. Building user engagement and trust is essential in HMRI. Users need to feel confident in the robots' capabilities and the reliability of the overall system. This can be achieved through introductory information, transparent decision-making processes, and real-time feedback mechanisms [30, 40].

However, managing interactions between multiple robots and human users can be complex, requiring sophisticated algorithms and interfaces to ensure smooth collaboration [11, 50]. The cognitive load on human users can be significant, especially when interacting with multiple robots simultaneously. Strategies to reduce this load, such as intuitive interfaces or intelligent support agents, are essential [45, 57]. In addition, the system must be adaptable and flexible to accommodate varying levels of user expertise and dynamic task requirements. This includes the ability to adapt to new scenarios in real-time [40].

In this regard, real-time visualization can play a critical role in empowering non-expert users. It can provide users with immediate feedback on the status and progress of tasks, enabling them to make informed decisions and adjustments as needed. This can include visualization of the planned robot paths, the captured environment, and the overall workflow [48]. In addition, interactive visualization tools can increase user engagement by making the process more intuitive and enjoyable. Potentially leading to higher satisfaction and better results, especially for non-expert users [43]. The implementation of visualization can include augmented reality (AR), which can be used to overlay digital information on the physical environment to provide contextual information and guidance to the user. Such AR overlay can enhance the user's ability to interact with robots and understand the remanufacturing process [54, 61]. Interactive dashboards that display real-time data can help users monitor and control the remanufacturing process, and these dashboards can also be customized to display relevant information based on the user's

role and expertise [43]. Real-time simulation and modeling tools can help users visualize potential outcomes and make adjustments before implementing changes, empowering users to reduce errors and improve the overall efficiency of the remanufacturing process [13].

To empower non-experts, it is crucial to make such interfaces intuitive and easy to use. This includes minimizing the need for technical knowledge and providing clear instructions and feedback [25]. Providing adequate training and support, such as tutorials, help guides, and access to expert assistance when needed, can also help users become more confident and proficient in using the system. Allowing users to customize and personalize their interaction with the system can further increase their sense of ownership and control. This can include customizing the interface layout, selecting preferred visualization tools, and setting personal preferences [55].

Here, somaesthetics' emphasis on cultivating bodily awareness and sensitivity can add another important angle to HMRI, suggesting that our tactile, physical interactions with technology can profoundly influence cognitive and affective processes [47]. In the context of fashion remanufacturing, this approach can encourage HMRI designers to consider not only the visual and operational aspects of real-time visualization but also how users physically engage with the systems - how their body movements, tactile feedback, and embodied sensations shape the interaction. By incorporating somatic principles, interfaces bear the potential to resonate more deeply with non-expert users, transforming the interaction into a holistic, sensorial experience that bridges the gap between digital command, physical creativity, and the materiality of textiles.

Empowering non-expert users in apparel remanufacturing can lead to several positive outcomes. By enabling more people to participate, apparel remanufacturing is likely to receive a wider range of attention, ideas and creativity, leading to more sustainable solutions and designs [24, 35]. Empowered users can make better use of available material resources, reducing the shortage of skilled personnel and the costs associated with remanufacturing. In addition, real-time visualization and effective HMRI could allow users to achieve manufacturing qualities that are difficult to achieve manually, resulting in aesthetically enhanced and longer-lasting second life cycle products [12]. Overall, the integration of HMRI and real-time visualization in fashion remanufacturing has significant potential to empower non-expert users and, by addressing this challenge, contribute to a more sustainable fashion industry.

3 Related Works

This section reviews related work in the areas of non-expert user empowerment, human-robot interaction, and real-time visualization, highlighting key contributions and identifying gaps that this research aims to address.

Recent research highlights the importance of user-centered design in empowering non-expert users. One approach involves fostering user engagement through incremental methods, transparent interfaces, and the integration of domain knowledge, underscoring the need for intuitive systems to increase trust and usability in non-expert contexts [51]. Another study explored strategies for enabling non-experts to engage with complex systems in creative fabrication spaces, emphasizing the critical role of accessible tools

and mediation by design researchers [17]. Computational methods have also demonstrated how 3D design tasks can be made accessible to non-experts through user-friendly, function-driven tools [56].

In human-multi-robot interaction, studies have shown that combining direct and indirect communication methods enhances operator trust and efficiency in collaborative settings [41]. Graphical interfaces play an important role in improving workload management and promoting transparency [42]. Emphasizing the need for efficient information distribution to maximize system performance, further research has introduced a multi-level programming model for optimal task allocation in multi-human multi-robot interaction [33]. Other work has proposed an affective workload allocation controller for multi-human multi-robot teams, demonstrating the importance of dynamic workload management based on user performance [27].

The role of immersive technologies has also been explored in the context of user engagement. Research has shown that virtual reality (VR) interfaces can significantly enhance interaction in fashion-related contexts and that VR applications can enable non-experts to perform complex tasks with minimal guidance [23, 37]. Data visualization platforms have similarly been shown to help non-experts make informed decisions, a concept applicable to improving transparency and decision-making in apparel remanufacturing [10].

Despite advances in human-multi-robot interaction, intuitive interfaces, and real-time visualization, the application of these technologies to apparel remanufacturing remains underexplored [16]. While HMRI research has demonstrated the benefits of adaptive workload distribution and transparent interfaces in industrial contexts, the unique challenges of working with textiles - such as their deformable and variable properties - make direct adaptation of these solutions difficult [49]. Similarly, technologies such as virtual reality and data visualization have been successfully used to engage non-expert users in other domains, but their potential to facilitate garment remanufacturing processes has yet to be explored. As outlined in the introduction, existing remanufacturing methods in the fashion industry are largely manual and require highly skilled labor, limiting scalability and accessibility [2, 14, 28]. In addition, as already mentioned, it is difficult to fully automate existing manual processes due to the complex behavior of textiles [19, 38]. Therefore, we see a need to rethink remanufacturing processes from the ground up, using collaborative automation and innovative technologies, such as electrospinning.

Electrospinning describes the process of spraying polymers within a high voltage field to create layers of nanofibers with spider web-like fabric quality on the target surface [59]. While past visions of personal, non-expert textile production of whole garments using electrospinning haven't lived up to their promise, these technologies have not yet been integrated into collaborative robotic systems designed for empowering non-expert users in garment repair and/or remanufacturing [44, 59].

This research addresses the aforementioned gaps by developing a user-centered, dual-robot system that integrates real-time visualization and intuitive interfaces specifically designed for non-expert users in garment remanufacturing, demonstrated using electrospinning. By focusing on this intersection of robotics and sustainable fashion, this research contributes to both fields while increasing

user engagement and effectiveness in remanufacturing processes and addressing pressing environmental and societal challenges.

4 Method

For this research project, we identified two core features of the LBR iiwa collaborative robot that set it apart: its exceptionally smooth hand guiding capability, which offers a user-friendly experience for those unfamiliar with robots, and a commander unit resulting from recent developments in KUKA's collaborative robots. Positioned adjacent to LBR iiwa's flange (the mounting interface at the robot's final axis), the commander unit can be seamlessly integrated into the hand guiding process. Equipped with a scroll wheel and two buttons, this unit transforms the robot from a conventional output device into an engaging input device as well (Figure 2). This dual functionality is critical for enabling users to intuitively interact with the robot and guide its movements without having to write complex code [8, 52].



Figure 2: LBR iiwa's in-built commander unit with scroll wheel and two buttons.

The challenge with hand guiding or "Teaching-by-Demonstration" - a process in which a human guides a robot through a task so that the robot can later replicate the demonstrated actions autonomously - is its efficiency in simple pick-and-place processes but inefficiency in processes requiring hundreds of positions or precise small-scale movements. Traditional programming methods can be time-consuming and prone to error, particularly in tasks that require high precision. To address this, we developed a flexible, parametric system that transforms minimal input data from the robot and its commander unit into complex toolpaths for the remanufacturing process. This system leverages advanced algorithms to interpret user inputs and generate precise, optimized paths for the robots to follow, ensuring high accuracy and efficiency [8, 9]. Another challenge is the LBR iiwa's limited reach of 760mm. To operate on a human scale, we linked it with another collaborative robot (cobot), allowing control of its flange position via the commander unit. This dual-robot setup extends the operational range and flexibility of the system, enabling it to handle larger and more complex tasks. To define a customized remanufacturing process, the user points the robot's flange at a mannequin and sets a few positions on the apparel. The user can adjust the area to be repaired based on the distance

between the robot's tool and the textile surface; larger distances cover larger areas, and closely spaced points merge into a single repair toolpath (Figure 3). Once all the areas of interest are defined, the robotic system orchestrates movements for both robots, each customized based on specific input data. The LBR iiwa positions itself optimally so that the LBR iiwa can effectively execute its programmed motions. This coordinated approach maximizes the efficiency and precision of the overall remanufacturing process, ensuring high-quality results.

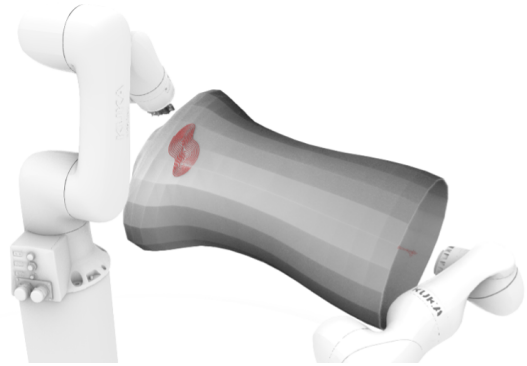


Figure 3: Simulation of two robot's collaborative task.

In the background, we use the visual programming environment Grasshopper for parametric toolpath planning and the visual live programming environment VVVV gamma for real-time control and interaction with the robot system [3, 15, 58]. The multi-robot toolpath planning is performed by KUKA|prc tools within the Grasshopper environment [4, 9]. The multi-robot real-time control, digital twin visualization and interaction are managed within VVVV gamma by the visual live programming tool Bunraku [5]. All relevant commands and information are visualized in real-time using the Stride 3D game engine implementation within VVVV gamma [58]. This integration provides users with a visualization of the entire process in a 3D environment, showing a comprehensive view of the robot's planned actions and the state of the remanufacturing process (Figure 4)[7].

For several reasons, real-time visualization is a critical component of our setup. First, it provides immediate feedback to users, allowing them to see the effects of their inputs and actions immediately. This immediate feedback loop ensures greater accuracy and reduces the likelihood of costly mistakes. Users can make adjustments on the fly, correcting errors before they escalate and optimizing the process for better results. Second, real-time visualization enhances user interaction by displaying a digital twin of the robot's workspace. This visual representation makes the technology accessible and easier to use, especially for users who may not have a background in robotics, allowing them to focus on creativity rather than technical details [31]. The digital twin mirrors the real-world setup, allowing users to intuitively understand the relationship between their actions and the robot's movements. This intuitive interface lowers the barrier to entry for non-experts, making advanced robotic technology more accessible and usable [21]. In addition, real-time visualization enables continuous process

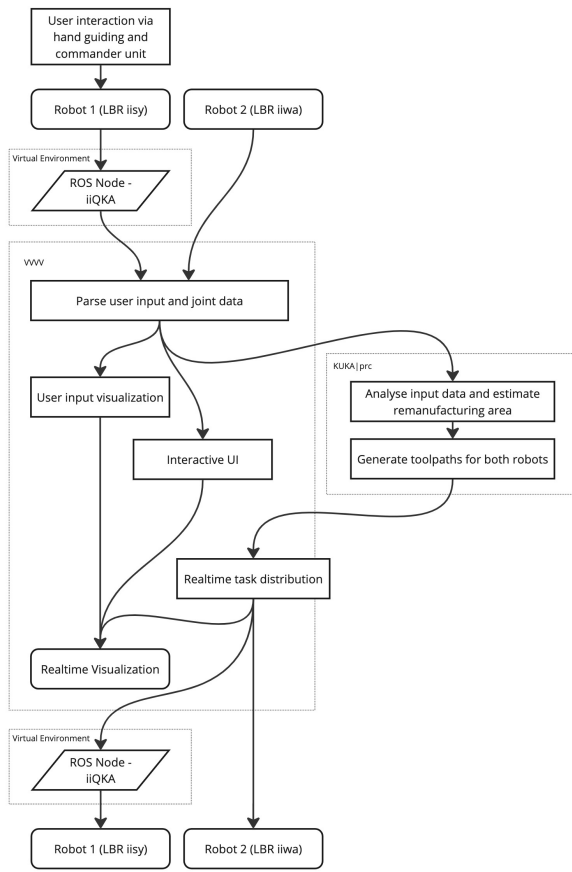


Figure 4: HMRI data pipeline overview.

monitoring, allowing users to observe the current position and movements of the robots, the status of toolpaths, and any changes in the textile being processed. This continuous monitoring is essential to ensure that the process meets the desired specifications and to identify any deviations or problems in a timely manner [29, 36]. Users can track the progress of the remanufacturing process in real-time, making it easier to ensure quality and consistency. Furthermore, it improves collaboration with multiple robots. Users can see how the robots interact with each other and with the textile, ensuring coordinated movements, which is critical for tasks requiring high precision [39]. Visualization helps plan and execute complex multi-robot tasks, ensuring that each robot's actions are harmonized for optimal performance [7]. Real-time visualization also supports customization and flexibility in the remanufacturing process. Users can dynamically adjust their input parameters and see the immediate impact of these changes, allowing the creation of highly customized remanufacturing processes tailored to specific textiles and designs [34, 53]. This increases the overall value and quality of the output, as users can fine-tune the process to meet specific requirements and preferences (Figure 5).



Figure 5: User interacting with the collaborative robot with real-time visualization in the background.

Finally, real-time visualization serves as a valuable learning tool for users as they move from basic operation to imagining their own robotic processes. It helps users understand how different inputs can be utilized to affect the robot's behavior, facilitating a deeper understanding of the potential of collaborative HMRI systems. Hopefully, this first-person experience will inspire users to imagine new remanufacturing processes that will further contribute to the advancement of fashion remanufacturing.

5 Project

In this paper, we have applied our method to an HMRI for non-expert apparel remanufacturing with electrospinning, which has been presented at various venues under the title "Exploring the future of individualized apparel (re)manufacturing - 3D-Electrospinning for Non-Woven Textile Remanufacturing".

As outlined in the previous chapters, our approach is to create a platform to rethink remanufacturing processes in a collaborative, automated manner with new technologies. In our vision, the method of electrospinning will be used to repair organic fiber garments with synthetic silk, chitin or collagen, and polyester garments with polymers recovered from non-repairable polyester garments. In line with this vision, we have developed an early proof of concept. The prototype demonstrates a process that uses high voltage to apply nanofibers from recycled polymers to reusable clothing.

We envisioned this facility being installed with solid safety protocols on the ground floor of a city center. Under supervision, local people could place damaged but remanufacturable garments on the mannequin held by the LBR iiwa robot, and then use the smaller co-bot to design the remanufacturing themselves.

In our demonstrator, system readiness for the remanufacturing design process is signaled to the user by a "Collaboration Mode" sign at the top left of the screen and yellow LED lights on the flange of the LBR iisy (Figure 6). In this mode, the user is invited to use the LBR iisy's hand guiding to point its electrospinning nozzle at the garment and to use its commander "Enter" button to save the projected location on the garment surface. Each time a new point is saved, the system automatically creates a new multi-robot production task, visualizes it on the production timeline at the bottom of the screen, and displays the associated toolpaths on the digital mannequin. Using LBR iisy's commander buttons, the user can either delete the created task and toolpath or continue with

the design process. Points on the back of the mannequin can be reached by using LBR iiwa's commander wheel to rotate LBR iiwa's axis 7. The size of the spray area is defined by the distance between the nozzle and the mannequin surface, with a greater distance resulting in a larger spray area. The shape of the spray area can be varied from circular to elliptical by changing the angle between the nozzle and the mannequin surface. In addition, the spray area can be increased by creating overlapping spray areas. Whenever there is no overlap, an isolated spray task is created. Once the user is satisfied with the design, it is the supervisor's responsibility to put the system into production mode (Figure 6).

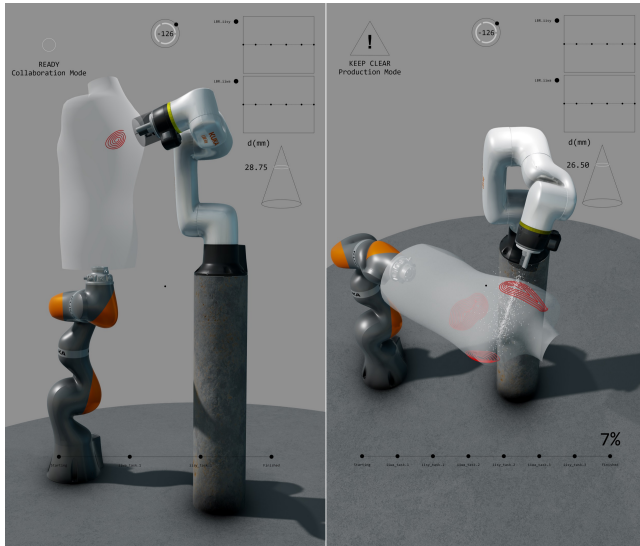


Figure 6: Real-time visualization screenshot of the collaboration mode (left side) and the production mode (right side).

In our lab we were able to demonstrate the full electrospinning remanufacturing process. For demonstrations in public venues this Production Mode involved a simulated multi-robot electrospinning process due to unavailable safety infrastructure (Figure 7). A video demonstrating the interaction behavior in more detail is available as supplementary material to this paper.

The project was presented as a proof-of-concept robotic system for fashion remanufacturing in semi-public venues such as automation trade shows and art and technology festivals. Visitors were invited to observe and interact with the system in a supervised environment. All interactions were part of a public exhibition with no identifiable information collected. We ensured that each demonstration adhered to ethical standards by explaining the purpose of the setup and obtaining implicit consent from participants who chose to engage. In doing so, our research followed the Austrian ethical guidelines for research and ensured no risks or privacy concerns for participants, as no sensitive or personal data was involved.

6 Preliminary Results and Learnings

This project was evaluated as a demonstrator without conducting a formal user study. As such, the findings are preliminary and require validation through planned user studies in future research.



Figure 7: Multi-robot system simulating the apparel remanufacturing with the electrospinning tool.

Designing the touch-based HMRI system around the robots' integrated commander interface and hand guiding capabilities improved system robustness. By eliminating the need for external input devices such as computer vision systems, this approach simplified the overall setup and reduced potential points of failure, ensuring a more reliable and streamlined interaction process.

The use of a high-resolution screen, as opposed to augmented reality glasses, proved to be an effective choice for reducing the barrier to entry for new users. After a short co-guided session of no more than 60 seconds, users were able to engage with the system and explore its advanced features on their own. This ease of use is particularly important for non-expert users who may be new to HMRI technology and/or AR.

The graphical user interface (UI) leverages the Stride rendering engine to provide high-quality, responsive rendering. This design was instrumental in increasing user satisfaction and engagement. The UI's intuitive and straightforward design minimized the learning curve, allowing users to quickly become proficient with the system.

Real-time visualization played a pivotal role in empowering users by providing immediate feedback on task status and progress. This feedback allowed users to make quick adjustments, reducing errors and avoiding delayed outcomes. By facilitating real-time monitoring and control of multiple robots, the system optimized workflow efficiency and reduced downtime. The combination of immediate feedback and clear task visualization seemed to significantly improve user engagement and decision-making.

Overall, the system's design encouraged intuitive interaction for non-expert users, resulting in a more seamless and efficient HMRI process. With robust graphical layers and additional safety measures, the system shows potential for operating autonomously without continuous supervision. However, due to the chemical and high-voltage components of electrospinning technology, the prototype presented in this paper required supervised operation.

The project also underscored the value of interdisciplinary collaboration, demonstrating how technologies from architecture and design can be applied to human-machine interaction scenarios. This cross-disciplinary approach fostered innovative solutions that improve both user experience and system efficiency.

In summary, this project highlights the critical role of robust, low-barrier and user-friendly interfaces in empowering non-expert users in multi-robot fashion remanufacturing. Preliminary results suggest that such can significantly improve user engagement, decision-making, and operational efficiency. However, future research must focus on validating these findings through a comprehensive user study.

7 Discussion

This section discusses the implications of our findings, the challenges we encountered, and the potential future directions for research and application in this domain.

The demonstrator highlights the importance of designing intuitive, low-barrier interfaces to empower non-expert users. The results confirm previous research that emphasizes incremental approaches and transparent interface design to build user trust and understanding. By following these principles, the demonstrator significantly increases user engagement and confidence, underscoring the value of accessible, user-centered design in complex robotic systems [51].

Real-time visualization proved to be a cornerstone in empowering non-expert users. Immediate feedback and dynamic task visualization enabled users to make informed decisions and adjustments during the redesign process. These findings are consistent with research advocating user experience (UX)-oriented approaches to human-robot encounters [43]. By extending these principles to fashion remanufacturing, the study demonstrates how visualization bridges the gap between human intuition and robotic precision, improving both usability and outcomes.

Effective task allocation and human-multi-robot coordination emerged as critical components. The demonstrator highlights the need for dynamic workload management and seamless communication between users and robots to optimize performance. These findings resonate with studies on workload allocation controllers and emphasize the importance of balancing human and robot contributions in collaborative tasks [27].

Building trust and ensuring safety have been fundamental to the success of HMRI. Transparent decision-making processes and real-time feedback mechanisms were shown to foster user trust. This is consistent with previous work emphasizing transparency as a key factor in multi-human, multi-robot systems [42]. By prioritizing these elements, a more user-friendly HMRI system tailored to the specific challenges of fashion remanufacturing is feasible.

Cognitive load management was identified as a significant challenge for non-expert users, especially when interacting with multiple robots simultaneously. The demonstrator integrates intuitive interfaces and real-time visualizations to mitigate this issue and balances transparency with ease of use to avoid out-of-the-loop performance issues [40]. These strategies align with research advocating human-in-the-loop approaches for collaborative human-robot teams, demonstrating how low-barrier tools can reduce complexity and improve user engagement [46].

While our demonstrator shows promising potential for empowering non-expert users in apparel remanufacturing, several scalability and practical implementation challenges remain. One significant barrier is the high cost and technical complexity inherent in integrating advanced collaborative robotic systems with high-voltage electrospinning tools, which also raises significant safety concerns in unsupervised settings. To mitigate these issues, future work could focus on modular, low-cost collaborative automation that allows for cost-effective scaling of garment remanufacturing. In addition, robust safety protocols covering both hardware and spatial redesign will be essential to ensure safe, unsupervised deployment.

In summary, the integration of HMRI and real-time visualization in fashion remanufacturing has the potential to empower non-expert users. By addressing the challenges and leveraging the benefits of these technologies, creativity, user satisfaction, and overall adoption of apparel remanufacturing can be enhanced. It is hoped that these findings will lay the groundwork for future research that explores the broader adoption and optimization of these technologies to promote sustainability and innovation in the fashion industry, paving the way for more accessible and effective fashion remanufacturing processes.

8 Outlook

Beyond the scope of this publication, our path of using visual programming environments to create HMRI fashion remanufacturing platforms as a next step aims to enable advancing users to quickly transition from users to developers who can further customize and optimize the presented remanufacturing processes or even develop their own from scratch. With this approach, we aim to democratize access to advanced robotics technologies further, allowing a broader range of users to engage with and benefit from these innovations. By fostering a deeper understanding and greater accessibility, we hope to drive the adoption of sustainable practices in the fashion industry and beyond.

Acknowledgments

The research for this paper was funded by the FWF PEEK Programme for Arts-based Research (project no.: AR611) and conducted by the Fashion and Robotics research team at the University of Art and Design Linz in collaboration with the Institute of Biomedical

Mechatronics, Prof. Werner Baumgartner, at the Johannes Kepler University Linz. The electrospinning process was refined in collaboration with Christoph Burgstaller and Harald Ladner from TCKT - Transfercenter für Kunststofftechnik GmbH. The electrospinning hardware was provided by Linari Engineering Srl as part of the Horizon project NanoWings (grant agreement no.: 101099620).

References

- [1] European Environment Agency. 2019. Textiles in Europe's circular economy. *Resource efficiency and waste* (2019), 11 pages. <https://doi.org/10.2800/904911>
- [2] Ranga Timur Aryaputra, Tangguh Okta Wibowo, Rennardo Juan Stefan, Ruth Yusiana, and Jessica. 2023. The Phenomenon of Thrifting As An Alternative Solution Related to Reducing Environmental Impact on Fast Fashion. *Formosa Journal of Multidisciplinary Research* 2, 8 (Aug. 2023), 1511–1526. <https://doi.org/10.55927/fjmr.v2i8.5696>
- [3] Robert McNeel & Associates. 2024. Rhino - Rhinoceros 3D. <https://www.rhino3d.com/>. [Accessed 23-09-2024].
- [4] Association for Robots in Architecture. 2024. Association for Robots in Architecture - making robots accessible to the creative industry. <https://robotsinarchitecture.org>. [Accessed 23-09-2024].
- [5] Amir Bastan. 2025. Bunraku. <https://amirbastan.com/projects/bunraku/>. [Accessed 08-02-2025].
- [6] Sophie Bernard. 2011. Remanufacturing. *Journal of Environmental Economics and Management* 62, 3 (2011), 337–351. <https://doi.org/10.1016/j.jeem.2011.05.005>
- [7] Johannes Braumann, Emanuel Gollob, and Amir Bastan. 2022. Towards AR for Large-Scale Robotics. In *2022 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*. IEEE, Christchurch, New Zealand, 15–18. <https://doi.org/10.1109/VRW55335.2022.00010>
- [8] Johannes Braumann and Karl Singline. 2021. Towards Real-Time Interaction with Industrial Robots in the Creative Industries. In *2021 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, Xi'an, China, 9453–9459. <https://doi.org/10.1109/ICRA48506.2021.9561024>
- [9] Johannes Braumann, Sven Stumm, and Sigrid Brell-Cokcan. 2016. Towards New Robotic Design Tools: Using Collaborative Robots within the Creative Industry. In *Proceedings of the 36th Annual Conference of the Association for Computer Aided Design in Architecture (ACADIA)*. ACADIA, Ann Arbor, USA, 164–173. <https://doi.org/10.52842/conf.acadia.2016.164>
- [10] Katrin Braunschweig, Julian Eberius, Maik Thiele, and Wolfgang Lehner. 2012. OPEN—Enabling Non-expert Users to Extract, Integrate, and Analyze Open Data. *Datenbank-Spektrum* 12 (2012), 121–130. <https://doi.org/10.1007/s13222-012-0091-9>
- [11] Jonathan Cacace, Alberto Finzi, and Vincenzo Lippiello. 2016. Implicit robot selection for human multi-robot interaction in Search and Rescue missions. *2016 25th IEEE International Symposium on Robot and Human Interactive Communication (RO-MAN)* (2016), 803–808. <https://doi.org/10.1109/ROMAN.2016.7745211>
- [12] Ling Chen, Zhi Su, Xiaotong He, Xiang Chen, and Lin Dong. 2022. The application of robotics and artificial intelligence in embroidery: challenges and benefits. *Assembly Automation* 42, 6 (2022), 851–868. <https://doi.org/10.1108/AA-07-2022-0183>
- [13] HaeSun Choi, Cindy Crump, Christian Duriez, Asher Elmquist, Gregory Hager, David Han, Frank Hearl, Jessica Hodgins, Abhinandan Jain, Frederick Leve, Chen Li, Franziska Meier, Dan Negrut, Ludovic Righetti, Alberto Rodriguez, Jie Tan, and Jeff Trinkle. 2021. On the use of simulation in robotics: Opportunities, challenges, and suggestions for moving forward. *Proceedings of the National Academy of Sciences* 118, 1 (2021), e1907856118. <https://doi.org/10.1073/pnas.1907856118>
- [14] Luz Claudio. 2007. Waste Couture: Environmental Impact of the Clothing Industry. *Environmental Health Perspectives* 115, 9 (2007), A449–A454. <https://doi.org/10.1289/ehp.115-a449>
- [15] Scott Davidson. 2024. Grasshopper - algorithmic modeling for Rhino. <https://www.grasshopper3d.com/>. [Accessed 23-09-2024].
- [16] Geetha Dissanayake and Pammi Sinha. 2015. An examination of the product development process for fashion remanufacturing. *Resources, Conservation and Recycling* 104 (2015), 94–102. <https://doi.org/10.1016/j.resconrec.2015.09.008>
- [17] Katrien Dreesen and Selina Schepers. 2020. Shifting towards community-building in opening up FabLabs for non-expert users. *Strategic Design Research Journal* 13 (2020), 24–41. <https://doi.org/10.4013/sdrj.2020.131.03>
- [18] Kaniz Farhana, Kumaran Kadirgama, Abu Shadate Faisal Mahamude, and Mushfika Tasnim Mica. 2022. Energy consumption, environmental impact, and implementation of renewable energy resources in global textile industries: an overview towards circularity and sustainability. *Materials Circular Economy* 4 (2022), 1–16. <https://doi.org/10.1007/s42824-022-00059-1>
- [19] Rocco Furferi, Yary Volpe, and Franco Mantellassi. 2022. Circular Economy Guidelines for the Textile Industry. *Sustainability* 14, 17 (2022), 20 pages. <https://doi.org/10.3390/su141711111>
- [20] Mosè Gallo. 2012. A Perspective on Remanufacturing Business: Issues and Opportunities. In *International Trade from Economic and Policy Perspective*, Vito Bobek (Ed.). IntechOpen, Rijeka, Chapter 10. <https://doi.org/10.5772/48103>
- [21] Emanuel Gollob, Magdalena Mayer, and Johannes Braumann. 2021. Using Robotics and A.I. to Physically Explore a Space of Aesthetic Possibilities: Defining a Physical Aesthetic Experience by the Targeted EEG Feedback of the Perceiver. In *Proceedings of the Fifteenth International Conference on Tangible, Embedded, and Embodied Interaction* (Salzburg, Austria) (TEI '21). Association for Computing Machinery, New York, NY, USA, Article 28, 8 pages. <https://doi.org/10.1145/3430524.3440647>
- [22] Carolina Gomes, Irina Pires, Leticia Monteiro, Tânia M. Lima, and Pedro D. Gaspar. 2024. Empowering Eco-Friendly Choices: An Environmental Impact Assessment Decision Support System for Textiles and Clothing. *Applied Sciences* 14, 2 (2024). <https://doi.org/10.3390/app14020659>
- [23] Uwe Gruenefeld, Tim Claudius Stratmann, Lars Prädel, Max Pfingsthorn, and Wilko Heuten. 2018. BuildingBlocks : Head-mounted Virtual Reality for Robot Interaction in Large Non-Expert Audiences. In *20th International Conference on Human-Computer Interaction with Mobile Devices and Services*. <https://api.semanticscholar.org/CorpusID:202741346>
- [24] Emily Huggard and Natalia Särämäki. 2023. How digital-only fashion brands are creating more participatory models of fashion co-design. *Fashion, Style & Popular Culture* 10, 4 (2023), 583–600. https://doi.org/10.1386/fspc.00176_1
- [25] Bryce Ikeda and Daniel Szafrir. 2024. PProgramAR: Augmented Reality End-User Robot Programming. *J. Hum.-Robot Interact.* 13, 1 (2024), 1 – 20. <https://doi.org/10.1145/3640008>
- [26] Wonse Jo, Jee Hwan Park, Sangjun Lee, Ahreum Lee, and Byung-Cheol Min. 2019. Design of a Human Multi-Robot Interaction Medium of Cognitive Perception. In *2019 14th ACM/IEEE International Conference on Human-Robot Interaction (HRI)*. 652–653. <https://doi.org/10.1109/HRI.2019.8673188>
- [27] Wonse Jo, Ruiqi Wang, Baijian Yang, Dan Foti, Mo Rastgaar, and Byung-Cheol Min. 2024. Cognitive Load-based Affective Workload Allocation for Multi-human Multi-robot Teams. <https://doi.org/10.48550/arXiv.2303.10465> arXiv:2303.10465
- [28] Siavash H. Khajavi. 2021. Additive Manufacturing in the Clothing Industry: Towards Sustainable New Business Models. *Applied Sciences* 11, 19 (2021). <https://doi.org/10.3390/app11198994>
- [29] Riccardo Karim Khamaisi, Elisa Prati, Margherita Peruzzini, Roberto Raffaeli, and Marcello Pellicciari. 2021. UX in AR-Supported Industrial Human–Robot Collaborative Tasks: A Systematic Review. *Applied Sciences* 11, 21 (2021). <https://doi.org/10.3390/app112110448>
- [30] Benedikt Leichtmann, Thomas Meneweger, Christine Busch, Bernhard Reiterer, Kathrin Meyer, Daniel Rammer, Roland Haring, and Martina Mara. 2024. Teaming with a Robot in Mixed Reality: Dynamics of Trust, Self-Efficacy, and Mental Models Affected by Information Richness. *International Journal of Human-Computer Interaction* 0, 0 (2024), 1–18. <https://doi.org/10.1080/10447318.2024.2331878>
- [31] Jessica Lindblom and Beatrice Alenljung. 2020. The ANEMONE: Theoretical Foundations for UX Evaluation of Action and Intention Recognition in Human-Robot Interaction. *Sensors* 20, 15 (2020). <https://doi.org/10.3390/s20154284>
- [32] Tal Madhala and William Fisher. 2023. Clothing, textile, and fashion industry sustainability impact measurement and management. *Acta IMEKO* 12, 4 (2023), 1–5. <https://doi.org/10.21014/actaimeko.v12i4.1336>
- [33] Monali S. Malvankar-Mehta and Siddhartha S. Mehta. 2015. Optimal task allocation in multi-human multi-robot interaction. *Optimization Letters* 9 (2015), 1787 – 1803. <https://doi.org/10.1007/s11590-015-0890-7>
- [34] Eloise Matheson, Riccardo Minto, Emanuele G. G. Zampieri, Maurizio Faccio, and Giulio Rosati. 2019. Human–Robot Collaboration in Manufacturing Applications: A Review. *Robotics* 8, 4 (2019). <https://doi.org/10.3390/robotics8040100>
- [35] Francesco Mazzarella, Helen Storey, and Dilys Williams. 2019. Counter-narratives Towards Sustainability in Fashion. Scoping an Academic Discourse on Fashion Activism through a Case Study on the Centre for Sustainable Fashion. *The Design Journal* 22, sup1 (2019), 821–833. <https://doi.org/10.1080/14606925.2019.1595402>
- [36] Fatemeh Mohammadi Amin, Maryam Rezayati, Hans Wernher van de Venn, and Hossein Karimpour. 2020. A Mixed-Perception Approach for Safe Human–Robot Collaboration in Industrial Automation. *Sensors* 20, 21 (2020). <https://doi.org/10.3390/s20216347>
- [37] Elena Morotti, Lorenzo Donatiello, and Gustavo Marfia. 2020. Fostering fashion retail experiences through virtual reality and voice assistants. In *2020 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*. 338–342. <https://doi.org/10.1109/VRW50115.2020.00074>
- [38] Geraldo Cardoso de Oliveira Neto, Luiz Fernando Rodrigues Pinto, Dirceu de Silva, Flavio Luiz Rodrigues, Fabio Richard Flausino, and Douglas Eldo Pereira de Oliveira. 2023. Industry 4.0 Technologies Promote Micro-Level Circular Economy but Neglect Strong Sustainability in Textile Industry. *Sustainability* 15, 14 (2023). <https://doi.org/10.3390/su151411076>
- [39] Uqba Othman and Erfu Yang. 2023. Human–Robot Collaborations in Smart Manufacturing Environments: Review and Outlook. *Sensors* 23, 12 (2023). <https://doi.org/10.3390/s23125663>
- [40] Jayam Patel and Carlo Pincirol. 2020. Improving Human Performance Using Mixed Granularity of Control in Multi-Human Multi-Robot Interaction. In *2020 29th IEEE International Conference on Robot and Human Interactive Communication*

- (RO-MAN). 1135–1142. <https://doi.org/10.1109/RO-MAN47096.2020.9223553>
- [41] Jayam Patel, Tyagaraja Ramaswamy, Zhi Li, and Carlo Pinciroli. 2021. Direct and Indirect Communication in Multi-Human Multi-Robot Interaction. <https://doi.org/10.48550/arXiv.2102.00672> arXiv:2102.00672 [cs.RO]
- [42] Jayam Patel, Tyagaraja Ramaswamy, Zhi Li, and Carlo Pinciroli. 2021. Transparency in Multi-Human Multi-Robot Interaction. <https://doi.org/10.48550/arXiv.2101.10495> arXiv:2101.10495 [cs.RO]
- [43] Elisa Prati, Margherita Peruzzini, Marcello Pellicciari, and Roberto Raffaeli. 2021. How to include User eXperience in the design of Human-Robot Interaction. *Robotics and Computer-Integrated Manufacturing* 68 (2021), 102072. <https://doi.org/10.1016/j.rcim.2020.102072>
- [44] Seyed Abdolkarim Hosseini Ravandi, Mehdi Sadrjahani, Afsaneh Valipouri, Farzad Dabirian, and Frank K Ko. 2022. Recently developed electrospinning methods: a review. *Textile Research Journal* 92, 23–24 (2022), 5130–5145. <https://doi.org/10.1177/00405175211069880>
- [45] Ariel Rosenfeld, Noa Agmon, Oleg Maksimov, and Sarit Kraus. 2017. Intelligent agent supporting human–multi-robot team collaboration. *Artificial Intelligence* 252 (2017), 211–231. <https://doi.org/10.1016/j.artint.2017.08.005>
- [46] Shayan Shayesteh and Houtan Jebelli. 2022. Toward Human-in-the-Loop Construction Robotics: Understanding Workers' Response through Trust Measurement during Human-Robot Collaboration. In *Construction Research Congress 2022*. American Society of Civil Engineers (ASCE), United States, 631–639. <https://doi.org/10.1061/9780784483961.066>
- [47] Richard Shusterman. 2008. *Body Consciousness: A Philosophy of Mindfulness and Somaesthetics*. Cambridge University Press.
- [48] Grimaldo Silva, Khansa Rekik, Ali Kanso, and Leizer Schnitman. 2022. Multi-perspective human robot interaction through an augmented video interface supported by deep learning. In *2022 31st IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*. 1168–1173. <https://doi.org/10.1109/RO-MAN53752.2022.9900671>
- [49] Jagdeep Singh, Kyungeun Sung, Tim Cooper, Katherine West, and Oksana Mont. 2019. Challenges and opportunities for scaling up upcycling businesses – The case of textile and wood upcycling businesses in the UK. *Resources, Conservation and Recycling* 150 (2019), 104439. <https://doi.org/10.1016/j.resconrec.2019.104439>
- [50] Elizabeth Sklar, Simon Parsons, A. Tuna Ozgelen, Eric Schneider, Michael Costantino, and Susan L. Epstein. 2013. HRTeam: a framework to support research on human/multi-robot interaction. In *Proceedings of the 2013 International Conference on Autonomous Agents and Multi-Agent Systems* (St. Paul, MN, USA) (AAMAS '13). International Foundation for Autonomous Agents and Multiagent Systems, Richland, SC, 1409–1410.
- [51] Richard Stromer, Oskar Triebe, Chad Zanolco, and Ram Rajagopal. 2024. Designing forecasting software for forecast users: Empowering non-experts to create and understand their own forecasts. <https://doi.org/10.48550/arXiv.2404.14575> arXiv:2404.14575 [cs.HC]
- [52] Sven Stumm, Johannes Braumann, and Sigrid Brell-Cokcan. 2016. Human-Machine Interaction for Intuitive Programming of Assembly Tasks in Construction. *Procedia CIRP* 44 (2016), 269–274. <https://doi.org/10.1016/j.procir.2016.02.108> 6th CIRP Conference on Assembly Technologies and Systems (CATS).
- [53] Hang Su, Wen Qi, Jiahao Chen, Chenguang Yang, Juan Sandoval, and Med Amine Laribi. 2023. Recent advancements in multimodal human–robot interaction. *Frontiers in Neurobotics* 17 (2023). <https://doi.org/10.3389/fnbot.2023.1084000>
- [54] Ryo Suzuki, Adnan Karim, Tian Xia, Hooman Hedayati, and Nicolai Marquardt. 2022. Augmented Reality and Robotics: A Survey and Taxonomy for AR-enhanced Human-Robot Interaction and Robotic Interfaces. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) (CHI '22). Association for Computing Machinery, New York, NY, USA, Article 553, 33 pages. <https://doi.org/10.1145/3491102.3517719>
- [55] Manasi Swaminathan, Long-Jing Hsu, Min Min Thant, Kyrie Jig Amon, Anna S. Kim, Katherine M. Tsui, Selma Sabanović, David J. Crandall, and Weslie Khoo. 2024. If [YourName] Can Code, So Can You! End-User Robot Programming For Non-Experts. In *Companion of the 2024 ACM/IEEE International Conference on Human-Robot Interaction* (Boulder, CO, USA) (HRI '24). Association for Computing Machinery, New York, NY, USA, 1033–1037. <https://doi.org/10.1145/3610978.3640644>
- [56] Nurcan Gecer Ulu. 2018. Computational Design and Evaluation Methods for Empowering Non-Experts in Digital Fabrication. (6 2018). <https://doi.org/10.1184/R1/6715244.v1>
- [57] Valeria Villani, Fabio Pini, Francesco Leali, and Cristian Secchi. 2018. Survey on human–robot collaboration in industrial settings: Safety, intuitive interfaces and applications. *Mechatronics* 55 (2018), 248–266. <https://doi.org/10.1016/j.mechatronics.2018.02.009>
- [58] vvvv group. 2024. vvvv - visual live-programming for .NET. <https://visualprogramming.net>. [Accessed 23-09-2024].
- [59] Joseph White, Marcus Foley, and Aaron Rowley. 2015. A Novel Approach to 3D-Printed Fabrics and Garments. In *3D Printing and Additive Manufacturing*, Vol. 2. 145–149. <https://doi.org/10.1089/3dp.2015.0019>
- [60] Linggang Zhang and Richard Vaughan. 2016. Optimal gaze-based robot selection in multi-human multi-robot interaction. In *2016 11th ACM/IEEE International Conference on Human-Robot Interaction (HRI)*. 645–646. <https://doi.org/10.1109/HRI.2016.7451898>
- [61] Chelsea Zou, Yan Ding, Kishan Chandan, and Shiqi Zhang. 2024. A Decision Framework for AR, Dialogue and Eye Gaze to Enhance Human-Robot Collaboration. In *Companion of the 2024 ACM/IEEE International Conference on Human-Robot Interaction* (Boulder, CO, USA) (HRI '24). Association for Computing Machinery, New York, NY, USA, 1209–1213. <https://doi.org/10.1145/3610978.3640550>